Disagreement on Tail

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Belief on Tail Events

- Tail events rarely happen but do matter.
- Rietz (1988) and Barro (2006) demonstrate that rare events significantly contribute to the equity premium.
- Disastrous events also shape economic agents' lifetime expectations and behaviors (Malmendier and Nagel, 2011; Bernile, Bhagwat, Rau, 2016).
  - This implies that agents have dispersed beliefs on the probability of rare events.

Disagreement

- Investor disagreement (i.e., heterogeneous beliefs) affects asset prices, volatility, and trading volume (e.g., Miller 1977; Duffie and Heaton, 1996; Hong and Sraer 2016; Atmaz and Basak 2018).

- Theoretically, **disagreement on tail events** has important implications for stock valuation and risk-sharing (Chen, Joslin, Tran 2012).

- But, how to empirically measure disagreement on tail events?

Existing Measures on Tail Risk and Disagreement

- There is a large body of work on how to capture tail risk.
  - Options: The most famous example is the VIX. Backus, Chernov, and Martin (2011) employ options to infer disastrous events.
  - Cross-sectional returns: Kelly and Jiang (2014) use monthly firm-level price crashes to identify common fluctuations in tail risk among individual stocks.
  - News: Manela and Moreira (2017) use textual analysis to measure news-implied tail risk.
- There is also a substantial body of work on how to capture disagreement.
  - Forecast dispersion: Diether, Malloy, Scherbina (2002) use analysts' earning forecast dispersion
  - Trading volume: Garfinkel (2009) uses unexplained trading volume
  - Option open interest divergence: Ge, Lin, Pearson (2016) use option put-call open interest divergence

How to Measure Disagreement on Tail Events

- Problem: Conventional methods won't work here, due to these challenges—

  1. Tail events, by definition, rarely happen *ex post*; too few observations.

  2. Analysts do not regularly express opinions about tail events.

How to Measure Disagreement on Tail Events

- Problem: Conventional methods won't work here, due to these challenges—

  1. Tail events, by definition, rarely happen *ex post*; too few observations.

  2. Analysts do not regularly express opinions about tail events.

- Solution: Machine Learning $\Rightarrow$ Neural Network Model

Preview of Main Results

1. We use Neural Network Model (NNM) to construct firm-level disagreement on tail (DOT) in the equity market.

2. We find DOT is significantly related to asset pricing.

   a) DOT negatively predicts future stock returns. A long-short portfolio yields an average return of **1.07%** per month (t = 2.98) and an FF5 alpha of **0.86%** per month (t = 4.45).

   b) Further analysis shows that this negative predictive relation is consistent with a mispricing channel.

Data

- We use 94 firm characteristics from Gu, Kelly, and Xiu (2020)
- Stock return and industry data are from CRSP.
- We restrict our sample to common stocks listed in the NYSE, AMEX, and NASDAQ. Additionally, we cross-sectionally remove the bottom decile of stocks by market equity period-by-period.
- Our sample spans from January 1957 to December 2021.

Methods: a Road Map

- Heterogeneous beliefs are generated by various machine agents, similar to Bali et al. (2025).
- To capture investors' beliefs about tail events, we propose a statistical classification method (He, Lyu and Zhou, 2024).
- We use the standard deviation of forecasts on tail events by different investors (Diether, Malloy, Scherbina, 2002).

Methods: Differentiated Investors

- To model the belief of individual investor n, we assume all of the investors observe a public, complete, and identical information set $Z = (z_{i,t})$, where $z_{i,t} = (z_{i,t}^1, z_{i,t}^2, \cdots, z_{i,t}^K)'$ is a 94-dimensional vector representing 94 firm characteristics of stock i in month t.

- The divergence in beliefs $X_n = (x_{n,i,t})$ is generated by assigning different weights to each characteristic:

  $$x_{n,i,t} = g_n(z_{i,t}) = D_n z_{i,t}$$

  where D_n is a diagonal matrix of rank 94, with each diagonal element following:

  $$d_{j,j} \sim N(0, 1), \text{I.I.D.}$$

Methods: Learning Target

- In Gu, Kelly, and Xiu (2020), they consider a general conditional risk premium formulation:

$$E_t[R_{i,t+1}] = f(z_{i,t})$$

- To account for disagreement on extreme events, we modify their methods. Instead of focusing on regression, we approach it as a classification task. We define four events: Tail (T), Down (D), Up (U), and Surge (S), corresponding to monthly expected returns in $(-\infty$, -25%), [-25%, 0%), [0%, 25%), [25%, $+\infty$), respectively.

- We frame the general conditional risk premium in a classification format:

$$E_{n,t}[\mathbf{1}_{i,t+1}] = f_{n,t}(x_{n,i,t})$$

where $\mathbf{1}_{i,t+1} = (1_{R_{i,t+1}\in T}, 1_{R_{i,t+1}\in D}, 1_{R_{i,t+1}\in U}, 1_{R_{i,t+1}\in S})'$ is the indicator vector, whose expectation gives the probabilities for the 4 events.

Methods: Neural Network Model

- In this study, to accommodate the complexity of investors' belief formation process, we employ the Neural Network Model (NNM) to express $f_{n,t}(x_{n,i,t})$.

- NNM is a computational model inspired by the way biological neural networks process information in the human brain. It is often considered the most powerful tool in machine learning and is theoretically grounded as "universal approximator" for any smooth predictive relationship (Hornik et al., 1989; Cybenko, 1989)

- Our NNM is a four-class classification feedforward neural network model, with the model setup similar to Gu, Kelly, and Xiu (2020).

Methods: Neural Network Model

- Consider a NNM with one input layer, L hidden layer, and one output layer. Each hidden layer uses the ReLU (Rectified Linear Unit) activation function[1], and the output layer uses the softmax activation function[2], which is typically used for multi-class classification problems.

$$E_{n,t}[\mathbf{1}_{i,t+1}] = NN_{n,t}\left(x_{n,i,t}\right) = softmax\left(W^{(L+1)}a^{(L)} + b^{(L+1)}\right)$$

where:

$$a^{(l)} = ReLU(u^{(l)}), \quad u^{(l)} = W^{(l)}a^{(l-1)} + b^{(l)}, \quad a^{(0)} = x_{n,i,t}, \quad l = 1, 2, \ldots, L$$

[1]The ReLU activation function is defined as: $ReLU(u_i^{(l)}) = \max(0, u_i^{(l)})$.

[2]The softmax activation function is defined as: $softmax(u_i^{(L+1)}) = \frac{e^{u_i^{(L+1)}}}{\sum_{j=1}^{M} e^{u_j^{(L+1)}}}$, where $u_i^{(L+1)}$ is the i-th element of the unactivated input $\mathbf{u}^{(L+1)}$, and M is the total number of classes.

Methods: DOT

- For stock i and month t, after generating beliefs on T, D, U and S from 100 machine investors, we employ standard deviation to capture the disagreement on tail (DOT):

$$DOT_{i,t} = \sqrt{\frac{1}{N-1}\sum_{n=1}^{N}\left(E_{n,t}[1_{R_{i,t+1}\in T}] - E_t[1_{R_{i,t+1}\in T}]\right)^2}$$

  where $E_t[1_{R_{i,t+1}\in T}] = \frac{1}{N}\sum_{n=1}^{N} E_{n,t}[1_{R_{i,t+1}\in T}]$ represents the mean of machine investors' beliefs on the tail event probability.

Univariate Portfolio Sorts on DOT

	EW				VW			
	Low	5	High	H-L	Low	5	High	H-L
RET	0.81***	0.94***	-0.17	-0.98***	0.68***	0.62**	-0.38	-1.07***
t-stat	(5.38)	(3.90)	(-0.39)	(-2.59)	(4.81)	(2.54)	(-0.92)	(-2.98)
CAPM	0.41***	0.29**	-1.15***	-1.55***	0.24***	-0.12	-1.44***	-1.68***
t-stat	(4.03)	(2.31)	(-4.02)	(-4.93)	(3.07)	(-1.62)	(-6.05)	(-5.75)
FF3	0.28***	0.12**	-1.27***	-1.56***	0.20***	-0.13*	-1.43***	-1.62***
t-stat	(3.71)	(2.02)	(-6.50)	(-6.57)	(3.19)	(-1.89)	(-7.23)	(-7.01)
FF5	0.14**	0.07	-0.74***	-0.89***	-0.01	-0.06	-0.87***	-0.86***
t-stat	(2.23)	(1.21)	(-3.77)	(-4.19)	(-0.30)	(-0.82)	(-4.68)	(-4.45)
Sharpe	0.80	0.59	-0.06	-0.39	0.64	0.36	-0.13	-0.44

Bivariate Portfolio Sorts on ME and DOT

Panel A: Equal-Weighted Portfolios							
	Low	5	High	H-L	t-stat	FF5	t-stat
ME Low	0.96	0.86	-0.32	-1.28***	(-3.06)	-1.27***	(-3.89)
ME 2	1.01	1.09	-0.28	-1.28***	(-3.20)	-1.32***	(-5.03)
ME 3	0.88	0.83	-0.22	-1.09***	(-2.89)	-1.01***	(-4.82)
ME 4	0.80	1.03	-0.06	-0.85***	(-2.59)	-0.69***	(-4.30)
ME High	0.65	0.72	0.42	-0.23	(-0.71)	-0.05	(-0.31)
Panel B: Value-Weighted Portfolios							
	Low	5	High	H-L	t-stat	FF5	t-stat
ME Low	0.95	0.86	-0.37	-1.32***	(-3.08)	-1.30***	(-3.84)
ME 2	1.01	1.10	-0.28	-1.29***	(-3.22)	-1.32***	(-5.08)
ME 3	0.86	0.86	-0.18	-1.04***	(-2.72)	-0.95***	(-4.65)
ME 4	0.79	1.02	-0.07	-0.86***	(-2.61)	-0.68***	(-4.32)
ME High	0.62	0.64	0.45	-0.17	(-0.55)	0.12	(0.72)

Fama-MacBeth Regressions

	RET	RET	Industry-adj. RET	Industry-adj. RET
DOT	-0.35***	-0.29***	-0.35***	-0.29***
	(-4.36)	(-3.92)	(-4.53)	(-3.94)
BETA	0.14*	0.14**	0.15***	0.15***
	(1.89)	(2.04)	(2.77)	(3.19)
SIZE	-0.18***	-0.25***	-0.16***	-0.21***
	(-3.57)	(-4.75)	(-3.50)	(-4.73)
BM	0.07*	0.06*	0.07**	0.07**
	(1.86)	(1.74)	(2.48)	(2.39)
GP	0.13***	0.10***	0.10***	0.08***
	(3.76)	(3.05)	(3.41)	(2.61)
AG	-0.22***	-0.21***	-0.20***	-0.19***
	(-6.72)	(-6.73)	(-7.48)	(-7.42)
ILL		-0.12***		-0.11***
		(-4.03)		(-3.82)
TURN		-0.15***		-0.15***
		(-4.13)		(-4.23)
MOM		0.35***		0.31***
		(6.27)		(6.60)
Const	0.76***	0.76***	0.10	0.10
	(3.07)	(3.06)	(1.01)	(1.01)
No. Obs.	3315	3142	3315	3142
Adj. R^2	5.37%	6.61%	3.79%	4.72%

Mispricing Channel: Institutional Ownership Ratio (IOR)

DOT negatively correlates with IOR, and DOT mainly works on low IOR groups.

Panel A: Average IOR in DOT Decile Portfolio					
	Low	5	High	H-L	t-stat
IOR	0.45	0.46	0.25	-0.19***	(-14.05)

Panel B: Bivariate Portfolio Sort on IOR							
	Low	5	High	H-L	t-stat	FF5	t-stat
IOR Low	0.78	0.29	-1.33	-2.10***	(-4.12)	-1.76***	(-5.14)
IOR 2	0.78	1.16	-0.03	-0.81	(-1.46)	-0.61*	(-1.79)
IOR 3	0.76	0.68	0.09	-0.66	(-1.41)	-0.42	(-1.37)
IOR 4	0.91	0.88	0.10	-0.80*	(-1.82)	-0.59**	(-2.26)
IOR High	0.79	0.83	0.44	-0.35	(-0.82)	-0.27	(-1.02)

Mispricing Channel: Mispricing Score (MISP)

DOT positively correlates with MISP, and MISP moderates DOT's predictive effect.

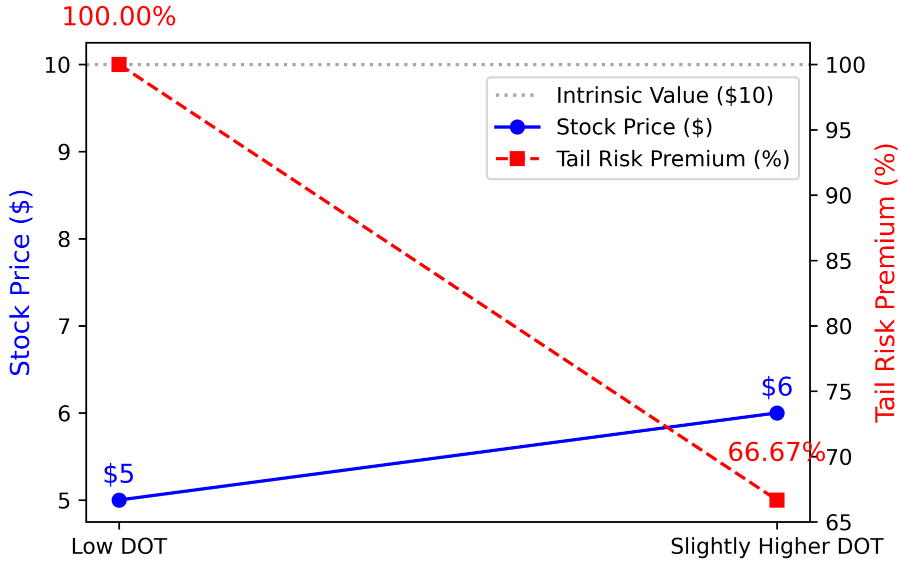
Panel A: Average MISP in DOT Decile Portfolio							
	Low	5	High	H-L	t-stat		
MISP	46.16	48.48	60.20	14.04***	(22.58)		
Panel B: Bivariate Portfolio Sort on MISP							
	Low	5	High	H-L	t-stat	FF5	t-stat
MISP Low	0.65	0.89	1.12	0.47	(1.48)	0.71***	(3.20)
MISP 2	0.60	0.80	0.74	0.13	(0.40)	0.49*	(1.66)
MISP 3	0.55	0.85	0.60	0.05	(0.15)	0.27	(0.98)
MISP 4	0.51	0.49	0.22	-0.29	(-0.65)	-0.08	(-0.28)
MISP High	0.30	0.06	-1.39	-1.69***	(-3.81)	-1.31***	(-4.62)

Mispricing Channel: Fama-MacBeth Regressions

The DOT-IOR and DOT-MISP interactions are significant.

	IOR sample			MISP Sample		
	RET	RET	RET	RET	RET	RET
DOT	-0.30***		-0.17*	-0.26***		-0.12**
	(-3.02)		(-1.82)	(-4.62)		(-2.30)
IOR		0.17***	0.12***			
		(3.72)	(2.76)			
DOT*IOR			0.18***			
			(4.91)			
MISP					-0.27***	-0.27***
					(-8.60)	(-8.53)
DOT*MISP						-0.12***
						(-5.64)
Const	0.80***	0.80***	0.85***	0.71***	0.71***	0.75***
	(3.00)	(3.00)	(3.14)	(2.89)	(2.90)	(3.06)
Controls	YES	YES	YES	YES	YES	YES
No. Obs.	3449	3449	3449	2554	2554	2554
Adj. R^2	6.15%	5.82%	6.48%	7.17%	6.99%	7.36%

Risk-Sharing Perspective: Illustration



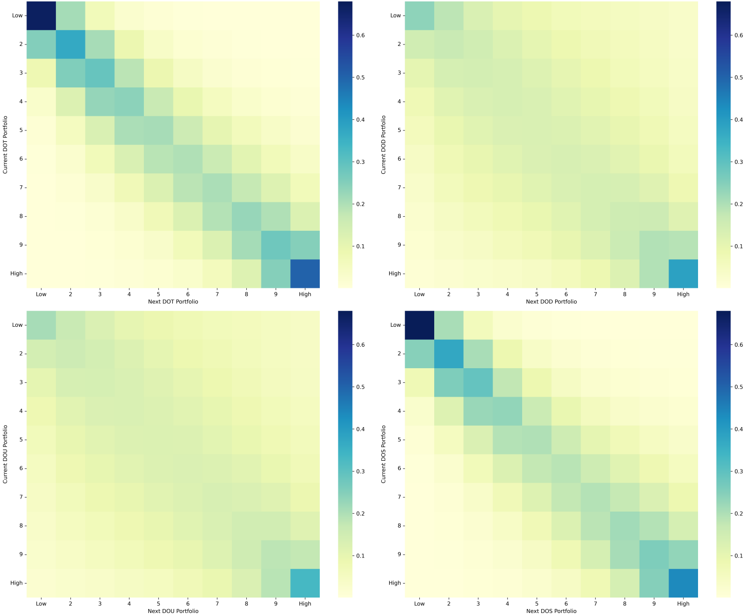
Risk-Sharing Perspective: Fama-MacBeth Regressions

DOT's predictive power more than doubles after tail events,
and the coefficient on I(DOT) becomes significant.

	Full Sample		After Tail Events	
	RET	RET	RET	RET
DOT	-0.30***		-0.78***	
	(-3.89)		(-2.97)	
I(DOT)		0.10		-0.89**
		(1.38)		(-2.14)
Const	0.74***	0.65***	2.43**	3.23***
	(3.02)	(2.94)	(2.56)	(3.84)
Controls	YES	YES	YES	YES
No. Obs.	3137	3137	281	281
Adj. R^2	6.53%	6.11%	5.63%	5.10%

I(DOT) = 1 if DOT $>$ 10th pct., 0 otherwise.

Transition Matrix of Disagreement on Different Events (from t-12 to t)

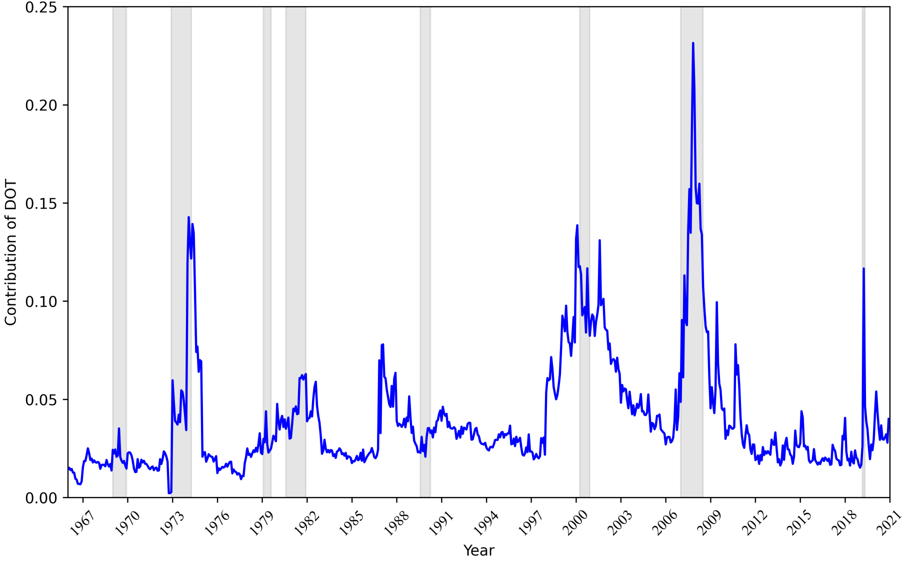


Univariate Portfolio Sorts on Disagreement on Different Events

DOT and DOS have predictive power for stock returns, whereas DOD and DOU do not.

	Low	5	High	H-L	t-stat	FF5	t-stat
DOT	0.68	0.74	-0.41	-1.09***	(-3.06)	-0.89***	(-5.16)
DOD	0.86	0.68	0.49	-0.37**	(-2.36)	-0.09	(-0.80)
DOU	0.77	0.76	0.54	-0.24	(-1.50)	-0.11	(-0.93)
DOS	0.66	0.79	0.02	-0.64**	(-2.07)	-0.53***	(-2.97)

The contribution of DOT to Total Disagreement



Conclusions

- We propose a statistical classification method to capture investors' heterogeneous beliefs about tail events (DOT).
- DOT negatively predicts future stock returns, consistent with a mispricing channel (Miller, 1977).
- The effectiveness of DOT is validated through the risk-sharing framework (Chen et al., 2012).
- The anatomy of disagreement reveals that DOT plays a crucial role, outweighing disagreement on other events.

Thank you!